

Mini Project

Employee Collection

Insert 50 records

Perform

Salary analysis

Department headcount

Top 5 highest paid employees

Create index on salary

Run aggregation pipeline

1. Create Employee Collection

Use CompanyDB

db.createCollection("Employees")

2. Insert 50 records

db.Employees.insertMany([

{ emp_name: "Arun Kumar", dept_name: "HR", salary: 35000 },
{ emp_name: "Meera Nair", dept_name: "Finance", salary: 42000 },
{ emp_name: "Rahul Das", dept_name: "IT", salary: 60000 },
{ emp_name: "Sneha Pillai", dept_name: "Marketing", salary: 45000 },
{ emp_name: "Vikram Singh", dept_name: "Sales", salary: 38000 },

{ emp_name: "Anjali Menon", dept_name: "HR", salary: 37000 },
{ emp_name: "Kiran Raj", dept_name: "IT", salary: 75000 },
{ emp_name: "Deepa N", dept_name: "Finance", salary: 50000 },
{ emp_name: "Aravind P", dept_name: "Sales", salary: 39000 },
{ emp_name: "Neethu S", dept_name: "Marketing", salary: 47000 },

{ emp_name: "Sanjay R", dept_name: "IT", salary: 82000 },
{ emp_name: "Priya K", dept_name: "HR", salary: 36000 },
{ emp_name: "Manoj V", dept_name: "Finance", salary: 53000 },
{ emp_name: "Divya L", dept_name: "Sales", salary: 40000 },
{ emp_name: "Ajith M", dept_name: "Marketing", salary: 48000 },

{ emp_name: "Rohit B", dept_name: "IT", salary: 90000 },
{ emp_name: "Lekshmi T", dept_name: "HR", salary: 34000 },
{ emp_name: "Nikhil S", dept_name: "Finance", salary: 55000 },
{ emp_name: "Fathima P", dept_name: "Sales", salary: 41000 },
{ emp_name: "Hari G", dept_name: "Marketing", salary: 46000 },

{ emp_name: "Gokul R", dept_name: "IT", salary: 78000 },

```

{ emp_name: "Athira M", dept_name: "HR", salary: 33000 },
{ emp_name: "Sreelakshmi K", dept_name: "Finance", salary: 52000 },
{ emp_name: "Rakesh P", dept_name: "Sales", salary: 42000 },
{ emp_name: "Anand S", dept_name: "Marketing", salary: 49000 },

{ emp_name: "Sharath V", dept_name: "IT", salary: 85000 },
{ emp_name: "Remya J", dept_name: "HR", salary: 36000 },
{ emp_name: "Naveen D", dept_name: "Finance", salary: 54000 },
{ emp_name: "Aswin T", dept_name: "Sales", salary: 43000 },
{ emp_name: "Bhavana R", dept_name: "Marketing", salary: 47000 },

{ emp_name: "Sidharth N", dept_name: "IT", salary: 88000 },
{ emp_name: "Anu P", dept_name: "HR", salary: 35000 },
{ emp_name: "Vishnu M", dept_name: "Finance", salary: 56000 },
{ emp_name: "Krishna L", dept_name: "Sales", salary: 41000 },
{ emp_name: "Akhil R", dept_name: "Marketing", salary: 50000 },

{ emp_name: "Varsha K", dept_name: "IT", salary: 92000 },
{ emp_name: "Reshma S", dept_name: "HR", salary: 34000 },
{ emp_name: "Ajay P", dept_name: "Finance", salary: 57000 },
{ emp_name: "Sreeraj T", dept_name: "Sales", salary: 44000 },
{ emp_name: "Nimisha B", dept_name: "Marketing", salary: 48000 },

{ emp_name: "Karthik R", dept_name: "IT", salary: 87000 },
{ emp_name: "Gayathri M", dept_name: "HR", salary: 36000 },
{ emp_name: "Albin P", dept_name: "Finance", salary: 59000 },
{ emp_name: "Jithin S", dept_name: "Sales", salary: 42000 },
{ emp_name: "Swathi L", dept_name: "Marketing", salary: 51000 },

{ emp_name: "Rahul K", dept_name: "IT", salary: 94000 },
{ emp_name: "Nayana V", dept_name: "HR", salary: 37000 },
{ emp_name: "Arjun M", dept_name: "Finance", salary: 60000 },
{ emp_name: "Vivek P", dept_name: "Sales", salary: 45000 },
{ emp_name: "Lakshmi R", dept_name: "Marketing", salary: 52000 }
])

```

3. Salary analysis

1.avg salary of all employees

```

db.Employees.aggregate({$group:{_id:null,average_salary:{$avg:"$salary"}}})
{
  _id: null,
  average_salary: 52400
}

```

2. Maximum and minimum salary

```
db.Employees.aggregate({$group: {_id: null, max_salary: {$max: "$salary"}, min_salary: {$min: "$salary"}}})
{
  _id: null,
  max_salary: 94000,
  min_salary: 33000
}
```

3. Total salary expense of the company

```
db.Employees.aggregate({$group: {_id: null, total_salary: {$sum: "$salary"}}})
{
  _id: null,
  total_salary: 2620000
}
```

4. department wise average salary

```
db.Employees.aggregate({$group: {_id: "$dept_name", avg_salary: {$avg: "$salary"}}})
{
  _id: 'Finance',
  avg_salary: 53800
}
{
  _id: 'IT',
  avg_salary: 83100
}
{
  _id: 'HR',
  avg_salary: 35300
}
{
  _id: 'Marketing',
  avg_salary: 48300
}
{
```



5 .department wise total salary

```
db.Employees.aggregate({$group: {_id: "$dept_name", avg_salary: {$sum: "$salary"}}})
{
  _id: 'IT',
  avg_salary: 831000
}
{
  _id: 'HR',
  avg_salary: 353000
}
{
  _id: 'Sales',
  avg_salary: 415000
}
```

Department HeadCount

```
db.Employees.aggregate({$group: {_id: "$dept_name", total_employees: {$sum: 1}}})
{
  _id: 'IT',
  total_employees: 10
}
{
  _id: 'HR',
  total_employees: 10
}
{
  _id: 'Sales',
  total_employees: 10
}
{
  _id: 'Marketing',
  total_employees: 10
}
```

Top 5 highest paid employees

```
> db.Employees.find().sort({salary:-1}).limit(5)
< [
  {
    _id: ObjectId('69a7b38b0198e3e3ba4a4c84'),
    emp_name: 'Rahul K',
    dept_name: 'IT',
    salary: 94000
  },
  {
    _id: ObjectId('69a7b38b0198e3e3ba4a4c7a'),
    emp_name: 'Varsha K',
    dept_name: 'IT',
    salary: 92000
  },
  {
    _id: ObjectId('69a7b38b0198e3e3ba4a4c66'),
    emp_name: 'Rohit B',
    dept_name: 'IT',
    salary: 90000
  }
]
```

Create index on salary

```
> db.Employees.createIndex({salary:1})
< salary_1
```

Run aggregation pipeline

```
db.Employees.aggregate([{$group:{_id:"$dept_name",avg_salary:{$avg:"$salary"}}})
[
  {
    _id: 'Finance',
    avg_salary: 53800
  },
  {
    _id: 'IT',
    avg_salary: 83100
  },
  {
    _id: 'HR',
    avg_salary: 35300
  },
  {
    _id: 'Marketing',
    avg_salary: 48300
  }
]
```

```
db.Employees.aggregate({$group:{_id:"$dept_name",avg_salary:{$sum:"$salary"}}})
{
  _id: 'IT',
  avg_salary: 831000
}
{
  _id: 'HR',
  avg_salary: 353000
}
{
  _id: 'Sales',
  avg_salary: 415000
}
```